AERIS Expert Review Packet

Anomaly

Source / metric	tceq / Nitrogen dioxide (no2)
Observed / expected baseline	11.1 / 1.32381 ppb
Baseline method	mean of the preceding 7 days of this series (Z-score detector)
Time	2026-07-02 02:00 UTC
Location	30.0395, -95.6739
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	Relative humidity (humidity)	percent	4 / 371	43.99 to 100; mean 73.4325	70.16 at 2026-07-02 01:55Z; 5 min before event
asos	Air temperature (temperature)	degC	4 / 371	22.7778 to 34.4444; mean 29.1171	29 at 2026-07-02 01:55Z; 5 min before event
asos	Wind direction (wind_direction)	degree	4 / 562	0 to 360; mean 120	150 at 2026-07-02 01:55Z; 5 min before event
asos	Wind speed (wind_speed)	m/s	4 / 575	0 to 11.8322; mean 2.9981	7.20222 at 2026-07-02 01:55Z; 5 min before event
noaa_gfs	500-hPa geopotential height (gh_500)	m	5 / 60	5894.32 to 5929.29; mean 5912.17	5902.41 at 2026-07-02 00:00Z; 2 h before event
noaa_gfs	Planetary boundary layer height (pbl_height)	m	5 / 60	34.8903 to 2463.44; mean 840.125	207.647 at 2026-07-02 00:00Z; 2 h before event
noaa_gfs	Precipitable water (precipitable_water)	mm	5 / 60	37.1521 to 56.5395; mean 45.5085	50.81 at 2026-07-02 00:00Z; 2 h before event
noaa_gfs	Surface pressure (surface_pressure)	hPa	5 / 60	1006.37 to 1013.78; mean 1011.07	1006.37 at 2026-07-02 00:00Z; 2 h before event
noaa_gfs	850-hPa temperature (t_850)	degC	5 / 60	17.459 to 19.5649; mean 18.6861	19.2318 at 2026-07-02 00:00Z; 2 h before event
noaa_gfs	10-m eastward wind (u_10m)	m/s	5 / 60	-4.80877 to 4.01123; mean -0.8825	-0.395796 at 2026-07-02 00:00Z; 2 h before event
noaa_gfs	10-m northward wind (v_10m)	m/s	5 / 60	-2.79096 to 5.8039; mean 2.1411	1.69125 at 2026-07-02 00:00Z; 2 h before event
openaq	Ozone (ozone)	ppm	7 / 454	0 to 0.065; mean 0.0192	0.008 at 2026-07-02 02:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
openaq	PM10 (pm10)	ug/m3	1 / 73	22 to 114; mean 47.6575	64 at 2026-07-02 02:00Z; at event time
openaq	PM2.5 (pm25)	ug/m3	60 / 2992	0 to 27.8; mean 5.4732	6.5 at 2026-07-02 02:00Z; at event time
openweather	Cloud cover (cloud_cover)	percent	3 / 217	0 to 100; mean 24.0553	21 at 2026-07-02 02:00Z; 1 min after event
openweather	Relative humidity (humidity)	percent	3 / 217	53 to 93; mean 74.765	71 at 2026-07-02 02:00Z; 1 min after event
openweather	Precipitation rate (precipitation)	mm/h	3 / 217	0 to 0; mean 0	0 at 2026-07-02 02:00Z; 1 min after event
openweather	Surface pressure (pressure)	hPa	3 / 217	1013 to 1017; mean 1015.4	1014 at 2026-07-02 02:00Z; 1 min after event
openweather	Air temperature (temperature)	degC	3 / 217	24.58 to 35.33; mean 29.9199	30.51 at 2026-07-02 02:00Z; 1 min after event
openweather	Wind direction (wind_direction)	degree	3 / 217	0 to 344; mean 148.931	153 at 2026-07-02 02:00Z; 1 min after event
openweather	Wind speed (wind_speed)	m/s	3 / 217	0.45 to 5.36; mean 1.8361	1.79 at 2026-07-02 02:00Z; 1 min after event
purpleair	PM2.5 (pm25)	ug/m3	42 / 2947	0 to 97.1; mean 6.8087	6.2 at 2026-07-02 01:59Z; at event time
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-01 19:07Z; 6 h 52 min before event
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-07-01 19:07Z; 6 h 52 min before event
sentinel5p	CO column (s5p_co_column)	mol/m^2	8 / 8	0.0261592 to 0.0295937; mean 0.0276	0.0269498 at 2026-07-01 19:07Z; 6 h 52 min before event
sentinel5p	CO granules available (s5p_co_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-01 19:07Z; 6 h 52 min before event
sentinel5p	HCHO column (s5p_hcho_column)	mol/m^2	8 / 8	0.000150572 to 0.000285154; mean 0.0002	0.000228552 at 2026-07-01 19:07Z; 6 h 52 min before event
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-01 19:07Z; 6 h 52 min before event
sentinel5p	NO2 column (s5p_no2_column)	mol/m^2	4 / 4	3.73882e-05 to 6.16311e-05; mean 0.0001	6.14335e-05 at 2026-07-01 19:07Z; 6 h 52 min before event
sentinel5p	NO2 granules available (s5p_no2_granule_available)	count	4 / 4	1 to 1; mean 1	1 at 2026-07-01 19:07Z; 6 h 52 min before event
sentinel5p	O3 granules available (s5p_o3_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-01 19:07Z; 6 h 52 min before event
sentinel5p	SO2 column (s5p_so2_column)	mol/m^2	8 / 8	-0.000214267 to 2.39858e-05; mean -0	-1.38712e-05 at 2026-07-01 19:07Z; 6 h 52 min before event
sentinel5p	SO2 granules available (s5p_so2_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-01 19:07Z; 6 h 52 min before event
tceq	Carbon monoxide (co)	ppm	1 / 65	0.1 to 0.9; mean 0.3954	0.3 at 2026-07-02 02:00Z; at event time
tceq	Nitrogen dioxide (no2)	ppb	6 / 375	0.5 to 26.9; mean 5.2293	11.1 at 2026-07-02 02:00Z; at event time

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	Relative humidity (humidity)	06-30 12Z 63.77, 18Z 55.27, 07-01 00Z 77, 06Z 91.6, 12Z 68, 18Z 54.58, 07-02 00Z 76.49, 06Z 91.58, 12Z 70.12, 18Z 61.04, 07-03 00Z 78.34, 06Z 90.14, 12Z 80.16
asos	Air temperature (temperature)	06-30 12Z 31.18, 18Z 32.87, 07-01 00Z 28.58, 06Z 26.1, 12Z 30.59, 18Z 32.76, 07-02 00Z 27.98, 06Z 25.66, 12Z 30.01, 18Z 31.18, 07-03 00Z 27.85, 06Z 25.24, 12Z 28.28
asos	Wind direction (wind_direction)	06-30 12Z 160, 18Z 148.5, 07-01 00Z 146.2, 06Z 118.8, 12Z 143.6, 18Z 149.1, 07-02 00Z 130.2, 06Z 71.88, 12Z 105, 18Z 124.8, 07-03 00Z 124.2, 06Z 54.68, 12Z 39.29
asos	Wind speed (wind_speed)	06-30 12Z 4.099, 18Z 5.562, 07-01 00Z 4.534, 06Z 1.972, 12Z 2.54, 18Z 4.341, 07-02 00Z 3.172, 06Z 1.072, 12Z 1.64, 18Z 4.619, 07-03 00Z 2.84, 06Z 0.6645, 12Z 0.7202
noaa_gfs	500-hPa geopotential height (gh_500)	06-30 18Z 5929, 07-01 00Z 5920, 06Z 5922, 12Z 5907, 18Z 5919, 07-02 00Z 5902, 06Z 5907, 12Z 5895, 18Z 5908, 07-03 00Z 5904, 06Z 5919, 12Z 5914
noaa_gfs	Planetary boundary layer height (pbl_height)	06-30 18Z 2345, 07-01 00Z 1134, 06Z 619.9, 12Z 142.1, 18Z 2192, 07-02 00Z 959.9, 06Z 304.6, 12Z 124.7, 18Z 1917, 07-03 00Z 185.6, 06Z 88.17, 12Z 68.94
noaa_gfs	Precipitable water (precipitable_water)	06-30 18Z 43.81, 07-01 00Z 43.12, 06Z 39.41, 12Z 38.27, 18Z 41.36, 07-02 00Z 47.82, 06Z 47.11, 12Z 49.31, 18Z 52.57, 07-03 00Z 54.6, 06Z 46.35, 12Z 42.37
noaa_gfs	Surface pressure (surface_pressure)	06-30 18Z 1012, 07-01 00Z 1010, 06Z 1011, 12Z 1011, 18Z 1011, 07-02 00Z 1009, 06Z 1011, 12Z 1011, 18Z 1012, 07-03 00Z 1011, 06Z 1012, 12Z 1012
noaa_gfs	850-hPa temperature (t_850)	06-30 18Z 18.46, 07-01 00Z 18.77, 06Z 19.47, 12Z 18.78, 18Z 18.4, 07-02 00Z 19.2, 06Z 18.85, 12Z 18.37, 18Z 17.62, 07-03 00Z 18.68, 06Z 18.86, 12Z 18.77
noaa_gfs	10-m eastward wind (u_10m)	06-30 18Z -1.067, 07-01 00Z -3.054, 06Z -1.626, 12Z -0.2487, 18Z -0.8201, 07-02 00Z -2.446, 06Z 0.07146, 12Z 0.5124, 18Z -1.019, 07-03 00Z -0.4448, 06Z -0.4377, 12Z -0.01089
noaa_gfs	10-m northward wind (v_10m)	06-30 18Z 3.396, 07-01 00Z 4.51, 06Z 3.156, 12Z 1.054, 18Z 2.52, 07-02 00Z 3.077, 06Z 2.343, 12Z 1.328, 18Z 1.638, 07-03 00Z -0.141, 06Z 1.746, 12Z 1.065
openaq	Ozone (ozone)	06-30 12Z 0.01908, 18Z 0.02863, 07-01 00Z 0.01689, 06Z 0.01221, 12Z 0.01695, 18Z 0.03511, 07-02 00Z 0.01779, 06Z 0.005743, 12Z 0.01317, 18Z 0.03788, 07-03 00Z 0.02129, 06Z 0.006543, 12Z 0.006619
openaq	PM10 (pm10)	06-30 12Z 46.75, 18Z 39.67, 07-01 00Z 34.17, 06Z 53.33, 12Z 50.5, 18Z 52.83, 07-02 00Z 60.33, 06Z 48.67, 12Z 70.33, 18Z 37.33, 07-03 00Z 36, 06Z 44.67, 12Z 41.67
openaq	PM2.5 (pm25)	06-30 12Z 5.546, 18Z 3.828, 07-01 00Z 5.468, 06Z 6.117, 12Z 6.551, 18Z 5.183, 07-02 00Z 5.292, 06Z 6.316, 12Z 6.113, 18Z 3.019, 07-03 00Z 4.564, 06Z 5.76, 12Z 6.357
openweather	Cloud cover (cloud_cover)	06-30 12Z 1.846, 18Z 3.833, 07-01 00Z 0.4118, 06Z 0.8947, 12Z 2.333, 18Z 8.389, 07-02 00Z 19.35, 06Z 37.74, 12Z 68.83, 18Z 85.06, 07-03 00Z 55.39, 06Z 4.778, 12Z 1.833
openweather	Relative humidity (humidity)	06-30 12Z 71.85, 18Z 61.17, 07-01 00Z 75.24, 06Z 87.47, 12Z 75.22, 18Z 57.61, 07-02 00Z 71.88, 06Z 86.79, 12Z 76.33, 18Z 63.11, 07-03 00Z 77.39, 06Z 86.39, 12Z 88
openweather	Precipitation rate (precipitation)	06-30 12Z 0, 18Z 0, 07-01 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-02 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-03 00Z 0, 06Z 0, 12Z 0
openweather	Surface pressure (pressure)	06-30 12Z 1017, 18Z 1015, 07-01 00Z 1015, 06Z 1015, 12Z 1016, 18Z 1014, 07-02 00Z 1014, 06Z 1015, 12Z 1016, 18Z 1015, 07-03 00Z 1015, 06Z 1016, 12Z 1017
openweather	Air temperature (temperature)	06-30 12Z 31.11, 18Z 33.98, 07-01 00Z 29.73, 06Z 26.91, 12Z 30.41, 18Z 34.61, 07-02 00Z 29.89, 06Z 26.48, 12Z 29.96, 18Z 33.06, 07-03 00Z 28.56, 06Z 26.06, 12Z 26.68
openweather	Wind direction (wind_direction)	06-30 12Z 139.5, 18Z 144.6, 07-01 00Z 123.1, 06Z 132.6, 12Z 169.9, 18Z 153.8, 07-02 00Z 143.7, 06Z 169.6, 12Z 139.3, 18Z 129.3, 07-03 00Z 165.1, 06Z 165.9, 12Z 167.5
openweather	Wind speed (wind_speed)	06-30 12Z 1.961, 18Z 2.955, 07-01 00Z 2.103, 06Z 1.386, 12Z 1.566, 18Z 2.484, 07-02 00Z 1.657, 06Z 1.428, 12Z 1.157, 18Z 2.648, 07-03 00Z 1.612, 06Z 1.466, 12Z 0.9267
purpleair	PM2.5 (pm25)	06-30 12Z 6.121, 18Z 4.907, 07-01 00Z 6.241, 06Z 7.424, 12Z 7.42, 18Z 5.826, 07-02 00Z 7.076, 06Z 8.757, 12Z 8.11, 18Z 4.793, 07-03 00Z 5.8, 06Z 8.188, 12Z 9.935
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	06-30 18Z 1, 07-01 18Z 1, 07-02 18Z 1

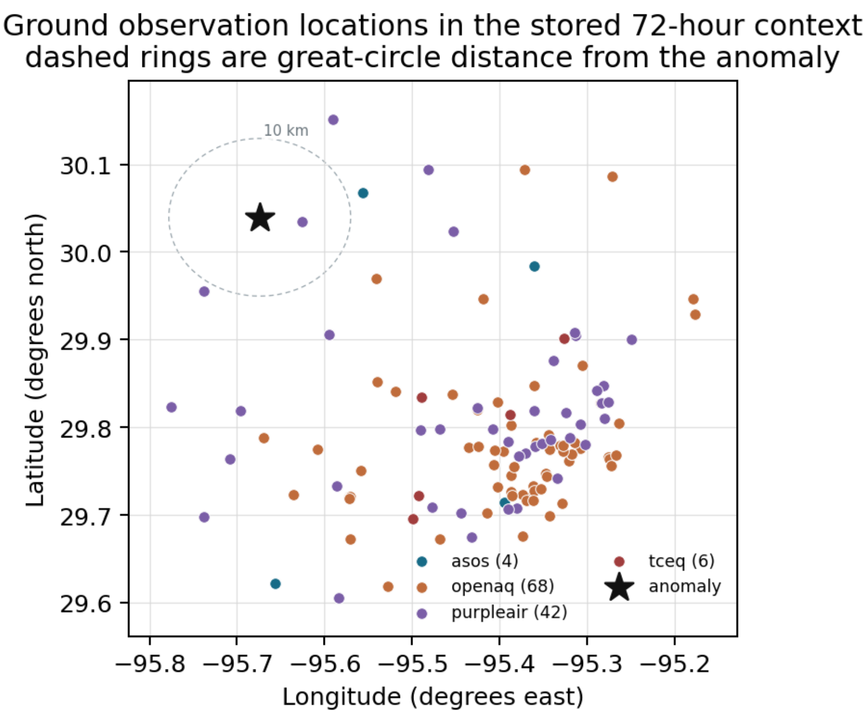
Source	Metric	Values
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	06-30 18Z 1, 07-01 18Z 1, 07-02 18Z 1
sentinel5p	CO column (s5p_co_column)	06-30 18Z 0.02626, 07-01 18Z 0.02689, 07-02 18Z 0.02856
sentinel5p	CO granules available (s5p_co_granule_available)	06-30 18Z 1, 07-01 18Z 1, 07-02 18Z 1
sentinel5p	HCHO column (s5p_hcho_column)	06-30 18Z 0.0001509, 07-01 18Z 0.0002309, 07-02 18Z 0.0002773
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	06-30 18Z 1, 07-01 18Z 1, 07-02 18Z 1
sentinel5p	NO2 column (s5p_no2_column)	06-30 18Z 3.739e-05, 07-01 18Z 6.143e-05, 07-02 18Z 5.589e-05
sentinel5p	NO2 granules available (s5p_no2_granule_available)	06-30 18Z 1, 07-01 18Z 1, 07-02 18Z 1
sentinel5p	O3 granules available (s5p_o3_granule_available)	06-30 18Z 1, 07-01 18Z 1, 07-02 18Z 1
sentinel5p	SO2 column (s5p_so2_column)	06-30 18Z 1.808e-05, 07-01 18Z 1.938e-06, 07-02 18Z -8.53e-05
sentinel5p	SO2 granules available (s5p_so2_granule_available)	06-30 18Z 1, 07-01 18Z 1, 07-02 18Z 1
tceq	Carbon monoxide (co)	06-30 12Z 0.35, 18Z 0.3167, 07-01 00Z 0.3667, 06Z 0.3667, 12Z 0.46, 18Z 0.3667, 07-02 00Z 0.375, 06Z 0.45, 12Z 0.4167, 18Z 0.4333, 07-03 00Z 0.425, 06Z 0.3333, 12Z 0.5333
tceq	Nitrogen dioxide (no2)	06-30 12Z 4.539, 18Z 3.256, 07-01 00Z 2.737, 06Z 5.064, 12Z 6.177, 18Z 4.642, 07-02 00Z 5.132, 06Z 5.586, 12Z 6.187, 18Z 6.755, 07-03 00Z 5.845, 06Z 5.728, 12Z 5.728

Per-site averages

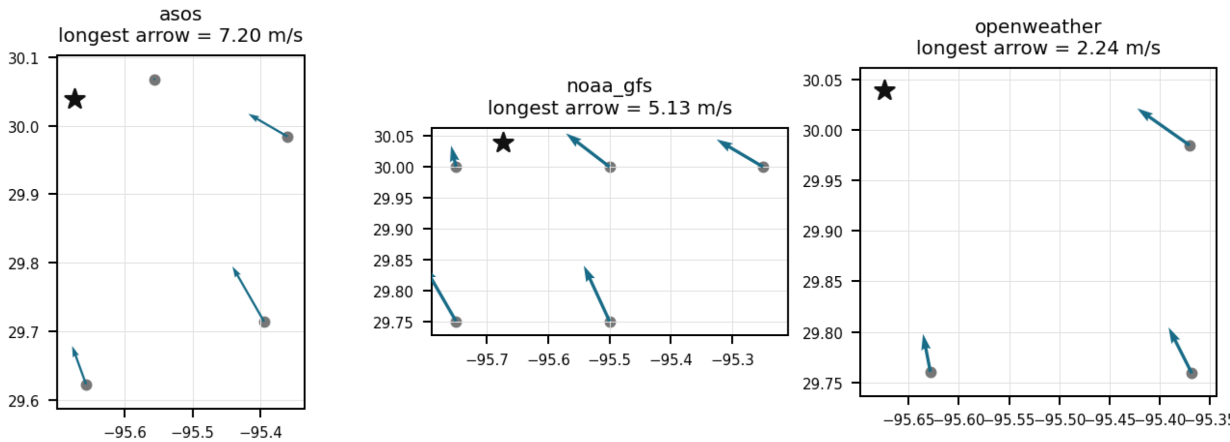
Each metric averaged at each site, with that site's distance from the event in brackets.

Source	Metric	Values
asos	Relative humidity (humidity)	DWH (11.767 km) 76.69, IAH (30.778 km) 72.53, MCJ (45.095 km) 71.1, SGR (46.424 km) 75.37
asos	Air temperature (temperature)	DWH (11.767 km) 28.09, IAH (30.778 km) 29.44, MCJ (45.095 km) 29.55, SGR (46.424 km) 29.06
asos	Wind direction (wind_direction)	DWH (11.767 km) 83.61, IAH (30.778 km) 126, MCJ (45.095 km) 137.3, SGR (46.424 km) 134.7
asos	Wind speed (wind_speed)	DWH (11.767 km) 1.354, IAH (30.778 km) 3.283, MCJ (45.095 km) 4.098, SGR (46.424 km) 3.259
noaa_gfs	500-hPa geopotential height (gh_500)	gfs:30.00,-95.75 (8.54 km) 5912, gfs:30.00,-95.50 (17.314 km) 5912, gfs:29.75,-95.75 (33.018 km) 5912, gfs:29.75,-95.50 (36.299 km) 5912, gfs:30.00,-95.25 (41.053 km) 5913
noaa_gfs	Planetary boundary layer height (pbl_height)	gfs:30.00,-95.75 (8.54 km) 653.8, gfs:30.00,-95.50 (17.314 km) 1062, gfs:29.75,-95.75 (33.018 km) 696.4, gfs:29.75,-95.50 (36.299 km) 928.7, gfs:30.00,-95.25 (41.053 km) 859.9
noaa_gfs	Precipitable water (precipitable_water)	gfs:30.00,-95.75 (8.54 km) 45.37, gfs:30.00,-95.50 (17.314 km) 45.84, gfs:29.75,-95.75 (33.018 km) 45.26, gfs:29.75,-95.50 (36.299 km) 44.9, gfs:30.00,-95.25 (41.053 km) 46.17
noaa_gfs	Surface pressure (surface_pressure)	gfs:30.00,-95.75 (8.54 km) 1009, gfs:30.00,-95.50 (17.314 km) 1011, gfs:29.75,-95.75 (33.018 km) 1011, gfs:29.75,-95.50 (36.299 km) 1012, gfs:30.00,-95.25 (41.053 km) 1012
noaa_gfs	850-hPa temperature (t_850)	gfs:30.00,-95.75 (8.54 km) 18.66, gfs:30.00,-95.50 (17.314 km) 18.68, gfs:29.75,-95.75 (33.018 km) 18.69, gfs:29.75,-95.50 (36.299 km) 18.73, gfs:30.00,-95.25 (41.053 km) 18.67
noaa_gfs	10-m eastward wind (u_10m)	gfs:30.00,-95.75 (8.54 km) -0.5441, gfs:30.00,-95.50 (17.314 km) -0.8033, gfs:29.75,-95.75 (33.018 km) -0.9633, gfs:29.75,-95.50 (36.299 km) -0.8733, gfs:30.00,-95.25 (41.053 km) -1.228
noaa_gfs	10-m northward wind (v_10m)	gfs:30.00,-95.75 (8.54 km) 2.53, gfs:30.00,-95.50 (17.314 km) 2.126, gfs:29.75,-95.75 (33.018 km) 2.315, gfs:29.75,-95.50 (36.299 km) 2.141, gfs:30.00,-95.25 (41.053 km) 1.595

Source	Metric	Values
openaq	Ozone (ozone)	3723 (0.015 km) 0.01989, 319 (28.951 km) 0.0182, 287 (35.354 km) 0.01817, 310 (36.873 km) 0.02229, 1319333 (39.972 km) 0.02147, 311 (41.77 km) 0.01713, 2046 (44.308 km) 0.01675
openaq	PM2.5 (pm25)	14140749 (15.005 km) 6.826, 14140750 (24.564 km) 6.753, 14157977 (26.633 km) 7.044, 10762809 (26.733 km) 7.129, 14140753 (27.9 km) 6.745, 14103803 (29.758 km) 4.301, 14152860 (30.129 km) 1.329, 12337668 (30.849 km) 6.902, +52 more
openweather	Cloud cover (cloud_cover)	grid:north (29.906 km) 25.15, grid:west (31.345 km) 23.81, grid:center (42.955 km) 23.22
openweather	Relative humidity (humidity)	grid:north (29.906 km) 75.35, grid:west (31.345 km) 74.29, grid:center (42.955 km) 74.66
openweather	Precipitation rate (precipitation)	grid:north (29.906 km) 0, grid:west (31.345 km) 0, grid:center (42.955 km) 0
openweather	Surface pressure (pressure)	grid:north (29.906 km) 1015, grid:west (31.345 km) 1015, grid:center (42.955 km) 1015
openweather	Air temperature (temperature)	grid:north (29.906 km) 29.72, grid:west (31.345 km) 29.78, grid:center (42.955 km) 30.26
openweather	Wind direction (wind_direction)	grid:north (29.906 km) 128.1, grid:west (31.345 km) 157.4, grid:center (42.955 km) 161.2
openweather	Wind speed (wind_speed)	grid:north (29.906 km) 1.992, grid:west (31.345 km) 1.598, grid:center (42.955 km) 1.917
purpleair	PM2.5 (pm25)	217323 (4.664 km) 7.654, 3033 (11.204 km) 5.182, 288282 (14.805 km) 9.782, 130763 (16.666 km) 2.203, 194469 (19.477 km) 10.2, 165203 (21.37 km) 5.454, 98139 (24.624 km) 7.854, 223797 (25.981 km) 6.417, +34 more
tceq	Nitrogen dioxide (no2)	482010029 (0.0 km) 3.157, 482010047 (28.955 km) 4.928, 482010024 (36.874 km) 4.33, 482011052 (37.243 km) 13.08, 482011066 (39.435 km) 2.821, 482010055 (41.774 km) 2.548



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Label definitions, from the labeling guide
Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it.
Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.
Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging.
Missing or thin evidence means Unsure, not Invalid.
A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.
Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict.

Claim 1 of 35

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5 particles.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 2 of 35

A fresh local combustion plume is supported because the TCEQ monitor at the anomaly site measured NO_2 at 11.1 ppb while nearby ozone was only 0.008 ppm at the same time, a pairing consistent with recently emitted NO titrating ozone before substantial downwind aging.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 3 of 35

A broad regional pollution or smoke episode is contradicted because particulate matter remained moderate near the event, Sentinel-5P CO was not unusually elevated before the event, and Sentinel-5P SO_2 showed no corresponding enhancement, which argues against a large-area transported combustion or sulfur-rich industrial plume as the dominant cause.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 4 of 35

A fresh nearby combustion emission plume, most plausibly from on-road traffic, diesel activity, industrial combustion, or power generation near the monitor, is a likely cause of the sharp NO_2 enhancement because the NO_2 spike occurred with simultaneously suppressed ozone, a pattern characteristic of recent NO titration in a young NO_x plume.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 5 of 35

Elevated regional tropospheric NO_2 column densities observed by Sentinel-5P satellite overpasses indicate urban-scale NO_x buildup across the Houston region prior to the surface event.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 6 of 35

The air quality anomaly in Houston, Texas is characterized by elevated levels of PM2.5 and NO2.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 7 of 35

Nocturnal boundary layer collapse and low surface wind speeds trapped local vehicular and industrial nitrogen oxide emissions near the ground, driving the surface NO2 anomaly detected by TCEQ monitors.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 8 of 35

NOAA GFS modeled planetary boundary layer height dropped significantly overnight alongside low ASOS surface wind speeds, supporting nocturnal boundary layer stagnation during the July 2 NO2 spike.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 9 of 35

Industrial activities and vehicle traffic are likely contributing factors to the air quality anomaly.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 10 of 35

Regional-scale alternative explanations are less supported because Sentinel-5P showed only modest NO2 enhancement on the prior daytime overpass, Sentinel-5P CO and SO2 did not show a strong regional plume, and surface PM2.5 stayed near typical single-digit micrograms per cubic meter levels around the event.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 11 of 35

The PM2.5 levels are elevated, with a mean of 6.8087 ug/m3 and a range of 0.0 ug/m3 to 97.1 ug/m3.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 12 of 35

The meteorological environment near 2026-07-02T02:00:00+00:00 featured humid air and modest south-southeasterly flow, and model guidance indicated a much shallower boundary layer later that night, which supports temporary near-surface trapping and poor dispersion of a local NOx plume.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 13 of 35

The anomaly occurred on July 2nd, around 02:00:38+00:00, with the nearest sensor reading 6.2 ug/m3 at a distance of 43.489 km from the anomaly location.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 14 of 35

A nearby industrial facility or power plant is releasing excessive amounts of particulate matter and volatile organic compounds into the atmosphere.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 15 of 35

Nighttime accumulation under a shallow and stabilizing boundary layer is a likely contributing cause of the elevated surface NO_2 because the event occurred near 02:00 UTC after sunset, with GFS boundary-layer height dropping from about 960 m at 00Z toward about 305 m by 06Z while humidity was high, conditions that commonly reduce vertical dilution of primary NO_x.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 16 of 35

The air quality anomaly is related to PM2.5 levels.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 17 of 35

The tceq NO2 measurement at 2026-07-02T02:00:00+00:00 at 30.040, -95.674 was 11.1 ppb, compared with an expected value of 1.323809523809524 ppb, yielding a z-score of 6.726124255396563 and a severity classification of severe.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 18 of 35

OpenAQ surface ozone measurements dropped to near-zero nighttime levels near 0.0057 ppm at the time of the TCEQ NO2 peak, supporting chemical titration of ozone by local emissions.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 19 of 35

Tropospheric NO2 column concentrations over the study area reached a peak 6-hour mean of 6.143e-05 mol/m^2 on 2026-07-01 at 18:00 UTC prior to the ground-level event.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 20 of 35

The nearest openaq ozone measurement at the anomaly time and location was 0.008 ppm at 2026-07-02T02:00:00+00:00, 0.015 km from 30.040, -95.674.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 21 of 35

The TCEQ NO2 monitor at 30.040, -95.674 measured 11.1 ppb at 2026-07-02T02:00:00+00:00, far above its expected value of 1.323809523809524 ppb, while nearby ozone was only 0.008 ppm at the same time, which is consistent with a fresh local NOx plume undergoing ozone titration.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 22 of 35

The air quality anomaly in Houston, Texas is caused by high levels of CO particles.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 23 of 35

The combination of stagnant air and high humidity is trapping pollutants near the surface, exacerbating the air quality anomaly.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 24 of 35

The anomaly is more consistent with a localized short-duration surface event than with a large regional smoke or photochemical episode because particulate matter and carbon monoxide were not simultaneously extreme, satellite SO2 was not elevated, and the same-period ozone near the site was low rather than high.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 25 of 35

NOAA GFS model output showed a shallow nocturnal boundary layer height dropping below 310 meters during the early morning hours of July 2, 2026, suppressing vertical atmospheric mixing.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 26 of 35

Nighttime accumulation under reduced mixing is supported because the anomaly occurred near local evening with high humidity around 70 to 76 percent and generally modest local winds near 1.8 to 3.2 m/s, while GFS boundary-layer height decreased from about 960 m at 00Z toward about 305 m by 06Z, indicating a transition toward a shallower nocturnal layer that could temporarily elevate surface NO2.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 27 of 35

Nocturnal titration of ambient ozone by fresh nitric oxide emissions converted NO into NO2 under suppressed vertical mixing, reflected by near-zero O3 levels recorded in OpenAQ ground data during the NO2 peak.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 28 of 35

Sentinel-5P NO2 total column density measurements peaked at 6.14e-5 mol/m^2 on July 1 before the surface event, providing evidence of regional tropospheric NOx accumulation across the Houston area.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 29 of 35

The combination of high PM2.5 and NO2 levels is a significant concern for public health in Houston, Texas.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 30 of 35

The air quality anomaly in Houston, Texas is caused by high levels of NO2 particles.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 31 of 35

A severe NO2 anomaly with a concentration of 11.1 ppb, representing a z-score of 6.73 above the expected value of 1.32 ppb, was detected at station 482010029 (30.040, -95.674) on 2026-07-02 at 02:00:00 UTC.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 32 of 35

The temperature inversion over the Houston area is causing a buildup of pollutants near the surface.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 33 of 35

TCEQ ground monitoring detected a severe nitrogen dioxide spike reaching 11.1 ppb at 02:00 UTC on July 2, 2026, representing a z-score of 6.73 above expected values.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 34 of 35

The pollutant with the detected anomaly is nitrogen dioxide (NO2) measured by tceq.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 35 of 35

Sentinel-5P satellite measurements recorded elevated tropospheric nitrogen dioxide column densities exceeding 5.5e-05 mol/m^2 over the Houston area during surrounding satellite passes.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.